SecureShield

Transparent GenAI Claim Assistant for Indian Health Insurance

Problem Statement

Indian health insurance customers face routine, unexpected claim denials due to opaque policy clauses (e.g., room-rent limits, "active line of treatment," "reasonable and customary" caps). Hospital staff, doctors, and patients struggle to interpret 30–50 page PDF policies under time pressure. This lack of upfront clarity leads to avoidable partial or full denials, eroding trust and increasing the administrative burden for all stakeholders in India's rapidly growing health insurance market.

Motivation

Rising health insurance uptake in India is hampered by low policy literacy, causing financial distress from unexpected deductions and wasting hospital time on avoidable claim denials. GenAI offers a solution: SecureShield will act as an **interpreter** to convert complex policy language into clear, consistent, and machine-executable rules. By providing real-time eligibility checks *before* admission or treatment, SecureShield aims to make Indian health insurance fairer and more predictable, reducing disputes and appeals.

Application

SecureShield is a GenAI assistant for Indian hospital billing teams, doctors, and patients, covering private retail and employer group policies for planned procedures.

Workflow:

1. **Input:** Billing staff enters insurer, policy, and basic admission details (room, stay, procedure).
2. **Response:** System gives instant, structured verdict (e.g., "Likely Approved") with a plain-language explanation citing the relevant policy condition.
3. **Support:** Staff can adjust inputs (e.g., change room type) to instantly see coverage changes, facilitating transparent pre-consent patient conversations.

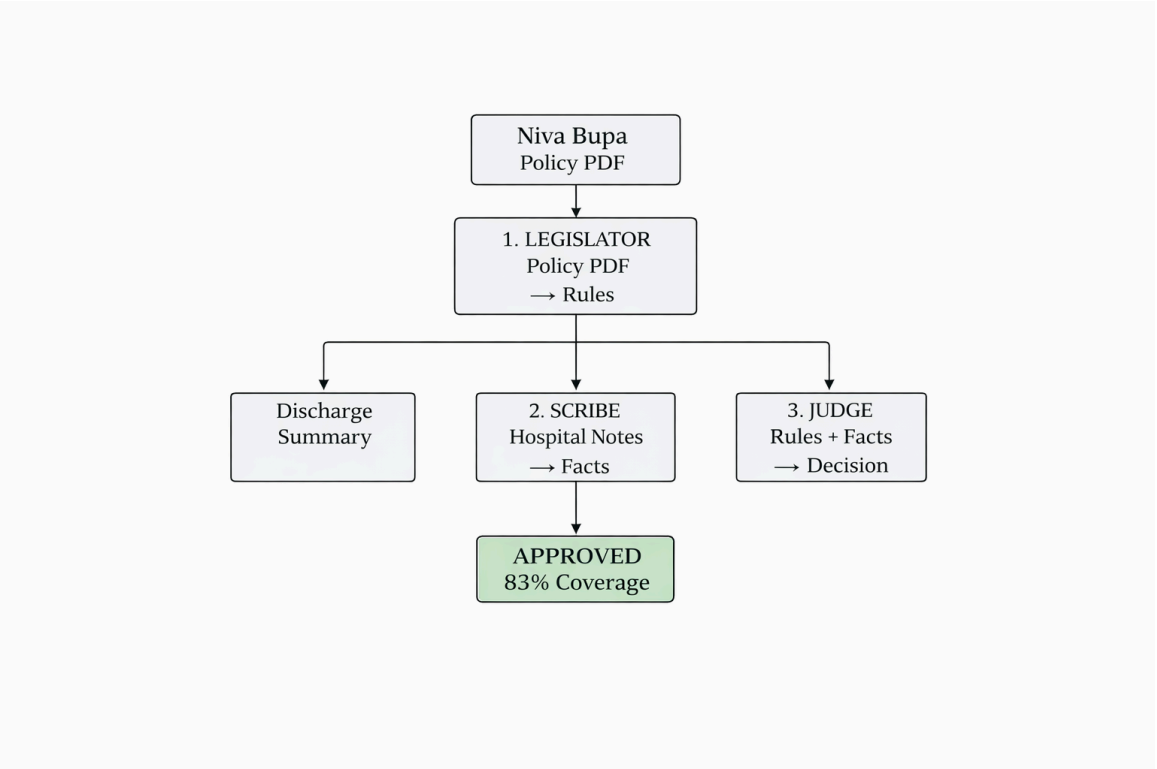
Future plans include integrating this interpretation into insurer portals and patient apps.

Proposed Method: Neuro-Symbolic Design:

This architecture ensures GenAI is used for language interpretation, while the final decision remains deterministic and explainable.SecureShield uses LLMs for language understanding and rule execution for reliable, auditable decisions.

1. **Policy Rule Extraction (Offline):** An LLM converts Indian health policy PDFs into structured, versioned, and frozen rules (JSON/tables), human-reviewed once.
2. **Case Fact Extraction (Online):** A model extracts key variables (e.g., duration, room cost) from structured inputs (forms) and unstructured text (notes) needed for the rules engine.
3. **Deterministic Decision Engine (Online):** A rules engine applies the *frozen* rules to the extracted *facts* to give an explainable verdict (approve/risk/deny) and levers for outcome change.

This ensures GenAI handles interpretation, while the final decision is deterministic and explainable.



Datesets / Data Source

The prototype uses public and synthetic data:

- **Policy documents:** Publicly available policy wordings and brochures from major Indian health insurers and IRDAI disclosures.
- **Regulatory guidelines:** IRDAI circulars on rules like room-rent caps and standard definitions.
- **Synthetic claim scenarios:** Manually designed test cases representing typical Indian hospitalizations with ground-truth outcomes labeled by policy analysis.

Experiments

Evaluation will occur in three layers:

- **Rule Correctness:** Compare system claim verdicts (approve/at-risk/deny) against ground-truth labels from manual policy reading, prioritizing avoiding false approvals.
- **Extraction Quality:** Assess field-level accuracy by comparing model-extracted variables (e.g., admission duration, room rent) from clinical descriptions against manually annotated values.
- **User Comprehension:** Confirm simulated staff/patients understand the system's explanations for "at-risk" claims and what specific change improves coverage.

Novelty and Scope to Scale

The core novelty is using GenAI as a **one-time translator** to convert policy text into stable, executable rules, rather than a black-box chatbot. This ensures reproducible, auditable outcomes, specifically tailored for Indian constructs like proportionate room-rent deduction, avoiding hallucinations.

The architecture is highly scalable, extendable from a few insurers to government schemes like PMJAY. Long-term, this explainable and fair approach can expand to other regulated, document-heavy Indian domains (e.g., loan underwriting, motor insurance).